

Assignment-D2

• Weather Forecasting

→ Weather Forecasting is something we all rely on from knowing if we need an umbrella tomorrow to prepare for serious events like hurricane or floods. Behind the scenes, it involves a lot of computing power and communication between different systems around the world.

• How Parallel computing helps

→ To predict the weather, scientists use complex computer models that simulate how the atmosphere behaves. These models involve tons of calculation just for money for a regular computer to handle quickly. That's where parallel computing comes in. Instead of using one processor to do all the work, parallel computing uses many processors at the same time to share the model. This helps meteorologists run their weather model faster and more accurately, which is super important when time is critical.

• How Networked system helps:

Weather data forms from all over the globe - satellites, airplanes, weather stations, and even ocean buoys. Networked systems are what connect all these resources, allowing the data to be sent in real time to weather centres. After the forecast is made, the same networks are used to send the information back out to news stations, weather apps, airports and emergency services.

Why it matters

Using both parallel computing and networked system together allows us to get reliable and timely weather updates. This helps people plan their day, Farmer protect their crops, airlines avoid dangerous condition and communication prepare for natural disaster.